Unnamed_Unit

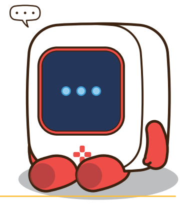
Unit 1 - Exploring the Core

PDF Documents

PDF Document 1: 1738067300971-Chapter 6_unit 1 - Exploring the Core.pdf

This PDF document is included in the unit materials.

UNIT 1 EXPLORING THE CORE



Topic Investigation

WHAT IS LIGHT?

Light is a form of energy known as electromagnetic radiation. It travels in waves and is part of a broad spectrum known as the electromagnetic spectrum. The light that we can see, known as visible light, is only a small part of this spectrum, which includes other types of waves like radio waves, infrared radiation, ultraviolet light, X-rays, and gamma rays.

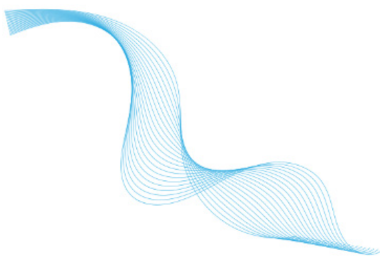
PROPERTIES OF LIGHT

SPEED

Light travels at an incredibly fast speed of approximately 299,792 kilometers per second in a vacuum. This speed changes slightly when light passes through different materials, like air or water.

WAVELENGTH AND FREQUENCY

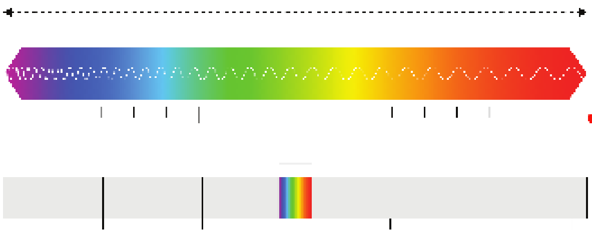
The wavelength of light determines its color. Blue light has shorter wavelengths, while red light has longer wavelengths. Frequency, which is inversely related to wavelength, refers to the number of wave cycles that pass a given point per second.

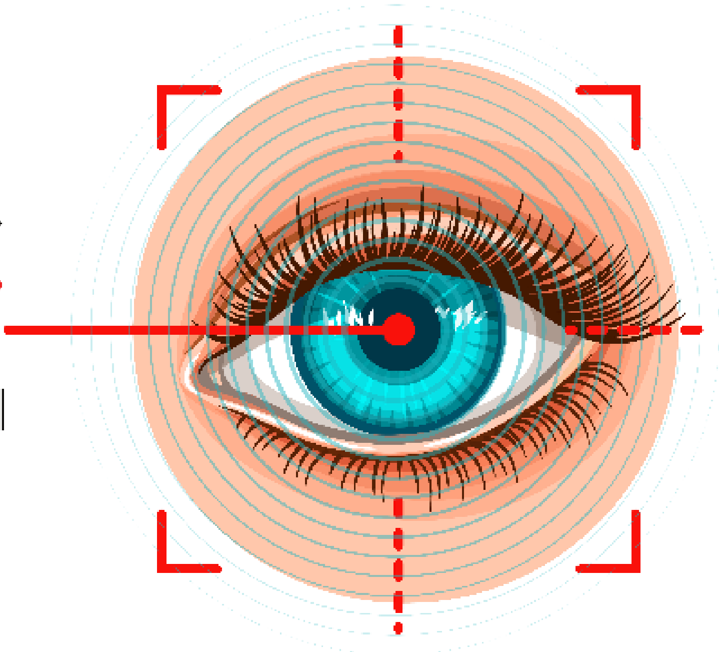


WAVE-PARTICLE DUALITY

Light exhibits both wave-like and particle-like properties. As a wave, it can interfere and diffract; as particles, called photons, light can be absorbed or emitted by substances.

HOW WE SEE LIGHT



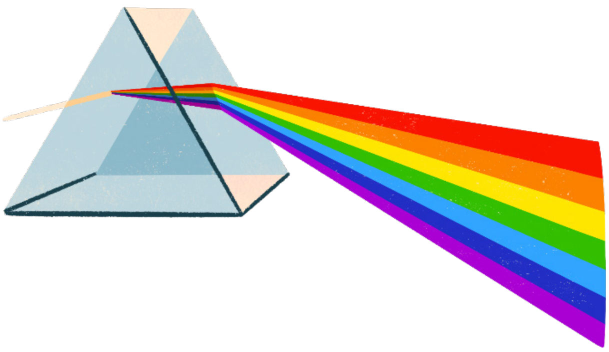


Our perception of light begins when it strikes an object and is either absorbed or reflected. The light that is reflected or transmitted reaches our eyes and is focused onto the retina, where it is converted into electrical signals that our brain interprets as images.

LIGHT INTERACTION WITH MATERIALS

Light can interact with different materials in various ways, including:

- **REFLECTION**
  Light bounces off the surface of a material.
- **REFRACTION:**
  Light bends as it passes through a material.
- **ABSORPTION:**
  Material absorbs light and converts it to other forms of energy
- **TRANSMISSION:**
  Light passes through a material without being absorbed.



Inquiry & Reflection

Discussion question 1

Why does a pencil look bent when it's half-submerged in a glass of water?

Discussion question 2

Why can we see different colors in soap bubbles?